Practice workbook · Deriving the double-angle formulae

Getting sin 2A, cos 2A (in all three forms) and tan 2A from the addition formulae by putting B = A.

24 question parts: 1 worked-example retry and 23 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-20

1. Question 11 · part a · 3 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part a only.

11 (a) Using a formula from page 2, show that $\cos 2x = 1 - 2\sin^2 x$

(3)

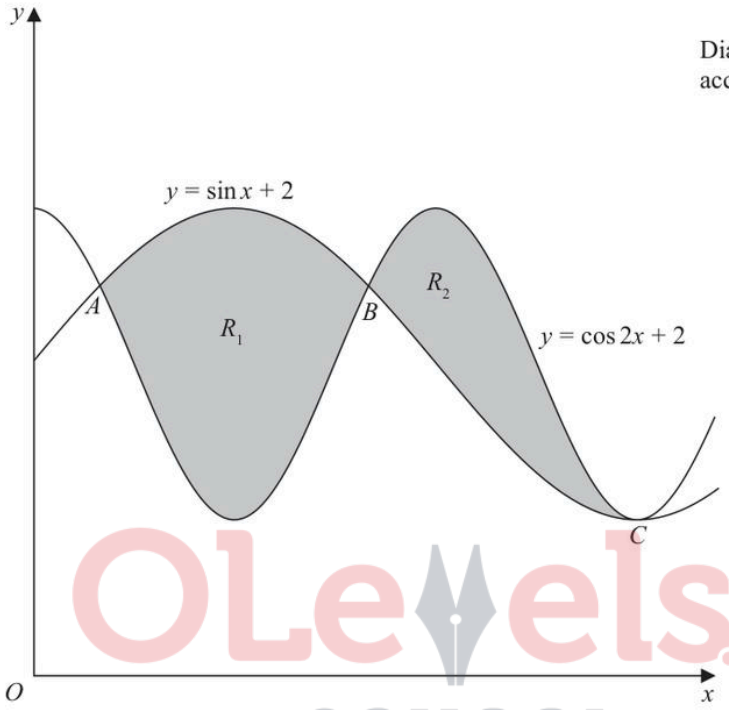


Figure 2

Figure 2 shows a sketch of part of the curves with equations $y = \sin x + 2$ and $y = \cos 2x + 2$

The points A, B and C, shown in Figure 2, are three points that are common to both curves.

(b) Find the coordinates of each of these points.

(4)

R_1 and R_2, shown shaded in Figure 2, are two regions enclosed by the two curves.

(c) Use calculus to find, in its simplest form, the ratio

area of R_1 : area of R_2

(8)

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1. Question 11 · part a · 3 marks · Working and answer

Working and answer

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2. Question 10 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

10 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Show that $\cos 4A = 8\cos^4 A - 8\cos^2 A + 1$

(4)

(c) Solve the equation $\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\left[\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right) - 1\right] = -\frac{1}{16} \qquad 0 \leqslant \theta < 2\pi$

Give your answers in terms of π.

(5)

$$f(A) = 4\cos^4 A - 4\cos^2 A + 1$$

(d) Using calculus, find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(A)\,dA$

Give your answer in the form $a\pi - b\sqrt{c}$ where a and b are fractions in their lowest terms and c is a prime number.

(4)

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2. Question 10 · part a · 2 marks · Working and answer

Working and answer

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3. Question 8 · part c · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

8 (a) Given that x is measured in radians, complete the table of values for

$$y = \sin x - 3\cos 2x - 0.5$$

Give your answers to one decimal place.

x	0	0.5	0.8	1	1.6	2	2.5	3
y			0.3	1.6			–0.8	–3.2

(2)

(b) On the grid opposite, draw the graph of $y = \sin x - 3\cos 2x - 0.5$ for $0 \leqslant x \leqslant 3$

(2)

(c) Use a formula from page 2 to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

$$f(x) = 2\sin x + 12\sin^2 x - x - 5$$

(d) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $f(x) = 0$ in the interval $0 \leqslant x \leqslant 3$

(5)

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3. Question 8 · part c · 2 marks · Working and answer

Working and answer

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4. Question 10 · part a.i · 0 marks

2023 JAN · Paper 02

Further practice · Attempt part a.i only.

10 Using formulae given on page 2

(a) show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

Given that $\theta \neq (90° + 180°n)$ where $n \in \mathbb{Z}$

(b) use the results from part (a) to show that $\sin 2\theta - \tan\theta$ can be written as $\tan\theta\cos 2\theta$

(4)

(c) Solve for $0 < x < 360$

$$\sin 2x° - \tan x° = 0$$

(4)

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4. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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5. Question 10 · part a.ii · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part a.ii only.

10 Using formulae given on page 2

(a) show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

Given that $\theta \neq (90° + 180°n)$ where $n \in \mathbb{Z}$

(b) use the results from part (a) to show that $\sin 2\theta - \tan\theta$ can be written as $\tan\theta\cos 2\theta$

(4)

(c) Solve for $0 < x < 360$

$$\sin 2x° - \tan x° = 0$$

(4)

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5. Question 10 · part a.ii · 5 marks · Working and answer

Working and answer

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6. Question 9 · part a.i · 0 marks

2023 JUN · Paper 01

Further practice · Attempt part a.i only.

9 (a) Using the formulae on page 2, show that

(i) $\cos^2 A = \dfrac{\cos 2A + 1}{2}$

(ii) $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

(4)

(b) Show that

$$(2\sin x - \cos x)(\sin x - 3\cos x) = \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$$

(5)

$$y = (2\sin x - \cos x)(\sin x - 3\cos x)$$

(c) Solve, for $0° \leqslant x \leqslant 180°$ the equation, $\dfrac{dy}{dx} = 0$

Give your answers to the nearest whole number.

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6. Question 9 · part a.i · 0 marks · Working and answer

Working and answer

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7. Question 9 · part a.ii · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part a.ii only.

9 (a) Using the formulae on page 2, show that

(i) $\cos^2 A = \dfrac{\cos 2A + 1}{2}$

(ii) $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

(4)

(b) Show that

$$(2\sin x - \cos x)(\sin x - 3\cos x) = \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$$

(5)

$$y = (2\sin x - \cos x)(\sin x - 3\cos x)$$

(c) Solve, for $0° \leqslant x \leqslant 180°$ the equation, $\dfrac{dy}{dx} = 0$

Give your answers to the nearest whole number.

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7. Question 9 · part a.ii · 4 marks · Working and answer

Working and answer

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8. Question 11 · part a · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

11 (a) Using a formula on page 2 show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Hence show that $\left(2\cos^2 A - 1\right)^2 = \dfrac{\cos 4A + 1}{2}$

(3)

The curve with equation $y = \dfrac{\sin 2x}{2} + \dfrac{\left(2\cos^2 x - 1\right)^2}{2} + \dfrac{1}{8}$ has a stationary point P in the range $0 \leqslant x \leqslant \dfrac{\pi}{6}$

(c) Find the exact coordinates of P

(7)

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8. Question 11 · part a · 2 marks · Working and answer

Working and answer

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9. Question 11 · part a.i · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a.i only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$

(3)

(ii) $\sin 2A = 2\sin A \cos A$

(1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$

(4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$

(4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$

(4)

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9. Question 11 · part a.i · 3 marks · Working and answer

Working and answer

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10. Question 11 · part a.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a.ii only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$ (3)

(ii) $\sin 2A = 2\sin A \cos A$ (1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ (4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$ (4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$ (4)

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10. Question 11 · part a.ii · 1 mark · Working and answer

Working and answer

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11. Question 8 · part a · 1 mark

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 (a) Write down the value of k such that $\sin 2A = k \sin A \cos A$

(1)

$$g(A) = 2 + 3\cos A - \sin A - 3\sin 2A - 2\cos^2 A$$

Given that $g(A)$ can be written in the form $(p\cos A - \sin A)(q - r\sin A)$ where p, q and r are integers,

(b) find the value of p, the value of q and the value of r.

(3)

(c) Hence solve, in radians to 3 significant figures where appropriate, the equation

$$g(2\theta) = 0 \quad \text{for} \quad 0 \leqslant \theta < \pi$$

(6)

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11. Question 8 · part a · 1 mark · Working and answer

Working and answer

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12. Question 1 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

1 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

Given that $\sin A = \frac{x+1}{2}$ and $\cos 2A = \frac{4-y}{3}$

(b) show that $y = \frac{1}{2}(3x^2 + 6x + 5)$

(3)

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12. Question 1 · part a · 2 marks · Working and answer

Working and answer

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13. Question 6 · part a · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

6 (a) Using a formula from page 2, show that $\cos 2A = 1 - 2\sin^2 A$

(2)

The finite region R is bounded by the curve with equation $y = 3 + 2\sin x$, the x-axis, the y-axis and the line with equation $x = \frac{\pi}{4}$

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.
Give your answer to the nearest integer.

(6)

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13. Question 6 · part a · 2 marks · Working and answer

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14. Question 7 · part a · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

7 (i) (a) Using a formula from page 2, show that

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

(2)

Given that $\tan 2\alpha = 1$

(b) show that $\tan\alpha = a \pm \sqrt{b}$ where a and b are integers whose values need to be found.

(3)

(ii) (a) Using formulae from page 2, show that $\cos(x-30)^\circ = \sin(x+30)^\circ$ can be written as $\tan x^\circ = 1$

(4)

(b) Hence, or otherwise, solve

$$\cos(2y-30)^\circ = \sin(2y+30)^\circ \qquad \text{for } -90 < y \leqslant 90$$

(2)

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14. Question 7 · part a · 2 marks · Working and answer

Working and answer

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15. Question 9 · part a · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part a only.

9 (a) Using a formula given on page 2, show that

$$\cos 2\theta = 2\cos^2\theta - 1$$

(2)

(b) Hence show that

$$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \left(2\cos^2\theta - 1\right) \mathrm{d}\theta = -\frac{a+\sqrt{b}}{c}$$

where a, b and c are integers to be found.

(4)

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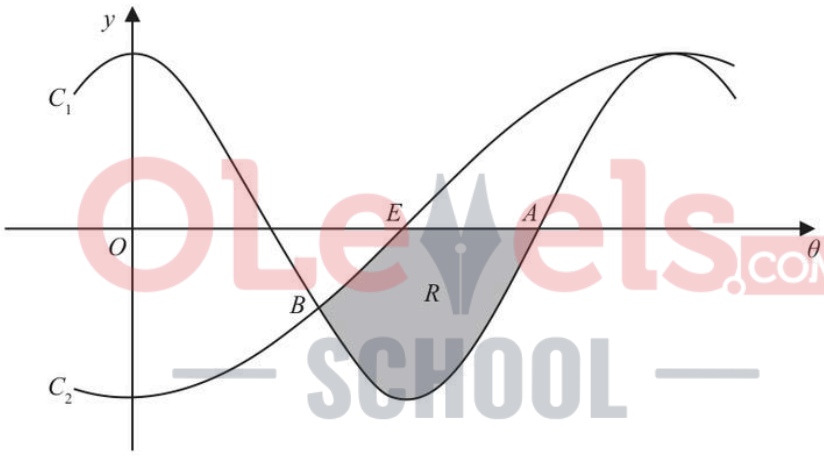


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 2\cos^2\theta - 1$ and part of the curve C_2 with equation $y = -\cos\theta$

Point B is the intersection of C_1 and C_2 as shown in Figure 3

Point $A\left(\frac{3\pi}{4}, 0\right)$ is the intersection of C_1 with the θ-axis as shown in Figure 3

Point $E\left(\frac{\pi}{2}, 0\right)$ is the intersection of C_2 with the θ-axis as shown in Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the θ-axis, C_1 and C_2

(c) Use calculus to find, in its simplest form, the exact area of R

(8)

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15. Question 9 · part a · 2 marks · Working and answer

Working and answer

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16. Question 11 · part a · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

11 (a) Use a formula on page 2 to show that $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$

(3)

(b) Show that $\sin^4 x + \cos^4 x = \frac{3 + \cos 4x}{4}$

(5)

(c) Hence solve, in degrees to one decimal place, the equation

$$8\sin^4\left(\frac{\theta}{2}\right) + 8\cos^4\left(\frac{\theta}{2}\right) = 5\sin(2\theta) + 6 \quad \text{for} \quad 0^\circ \leqslant \theta < 180^\circ$$

(4)

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16. Question 11 · part a · 3 marks · Working and answer

Working and answer

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17. Question 10 · part b.i · 0 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

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17. Question 10 · part b.i · 0 marks · Working and answer

Working and answer

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18. Question 10 · part b.ii · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

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18. Question 10 · part b.ii · 5 marks · Working and answer

Working and answer

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19. Question 10 · part a.i · 0 marks

2023 NOV · Paper 01

Further practice · Attempt part a.i only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

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19. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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20. Question 10 · part a.ii · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a.ii only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

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20. Question 10 · part a.ii · 3 marks · Working and answer

Working and answer

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21. Question 10 · part a.i · 0 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

10 (a) Using a formula from page 2, show that

(i) $\sin^2 A = \frac{1-\cos 2A}{2}$

(ii) $\cos^2 A = \frac{\cos 2A+1}{2}$

(5)

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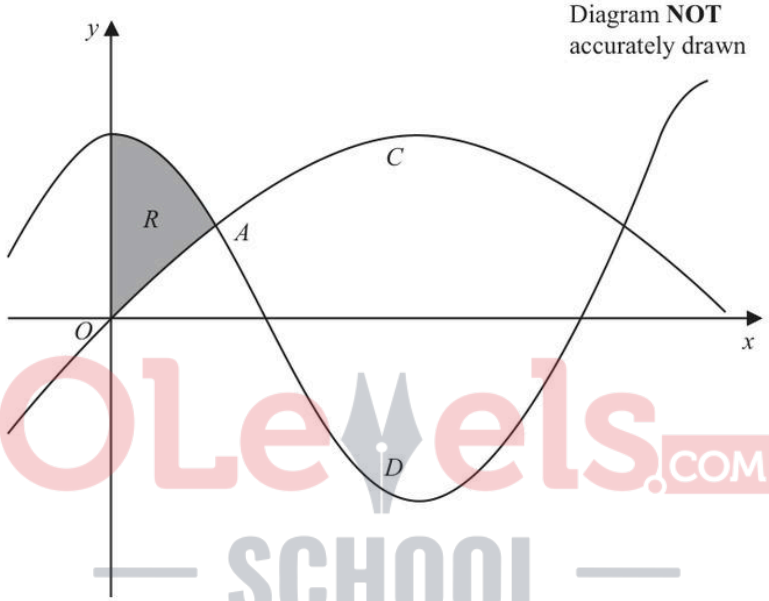


Figure 2

Figure 2 shows part of the curve C with equation $y = \sin x$ and part of the curve D with equation $y = \cos 2x$
Curve C and curve D intersect at the point A

(b) Use algebra to find the exact coordinates of A

(5)

The shaded region R is rotated through $360°$ about the x-axis.

(c) Use algebraic integration to find the exact volume of the solid generated.

Give your answer in the form $\frac{a\sqrt{a}}{b}\pi$

where a is a prime number and b is an integer.

(5)

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21. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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22. Question 10 · part a.ii · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

10 (a) Using a formula from page 2, show that

(i) $\sin^2 A = \frac{1-\cos 2A}{2}$

(ii) $\cos^2 A = \frac{\cos 2A+1}{2}$

(5)

Diagram **NOT** accurately drawn

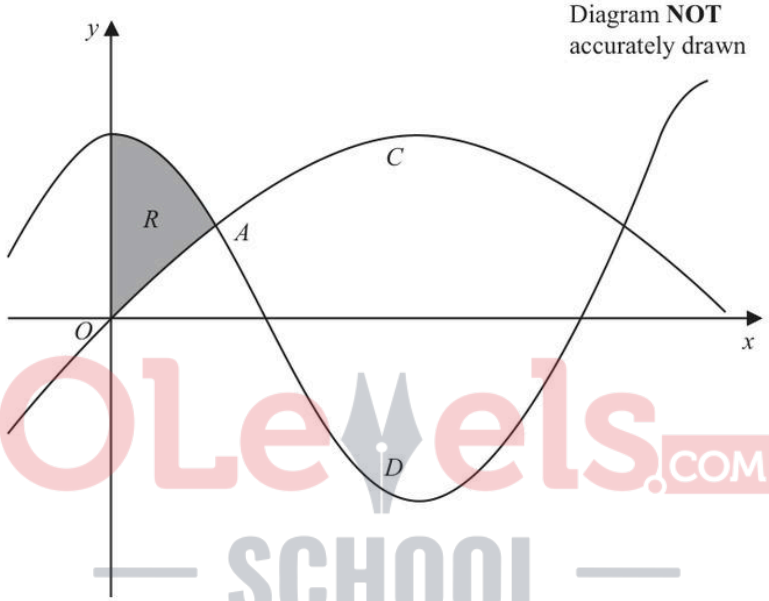


Figure 2

Figure 2 shows part of the curve C with equation $y = \sin x$ and part of the curve D with equation $y = \cos 2x$
Curve C and curve D intersect at the point A

(b) Use algebra to find the exact coordinates of A

(5)

The shaded region R is rotated through $360°$ about the x-axis.

(c) Use algebraic integration to find the exact volume of the solid generated.

Give your answer in the form $\frac{a\sqrt{a}}{b}\pi$

where a is a prime number and b is an integer.

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22. Question 10 · part a.ii · 5 marks · Working and answer

Working and answer

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23. Question 11 · part a · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

11 (a) Using a formula on page 2, show that

$$\frac{1-\cos 2A}{1+\cos 2A} = \tan^2 A$$

(3)

(b) Hence, or otherwise, solve in degrees to one decimal place

$$\frac{3-3\cos 4x}{1+\cos 4x} + \frac{5\sin 2x}{\cos 2x} = 2 \quad \text{for } -90^\circ < x < 90^\circ$$

(7)

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23. Question 11 · part a · 3 marks · Working and answer

Working and answer

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24. Question 8 · part a · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

8 (i) (a) Using a formula given on page 2, show that

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

(2)

(b) Hence, solve the equation

$$\tan A° - \tan 2A° = 0 \qquad \text{for } 0 \leqslant A \leqslant 180$$

(5)

(ii) Using a formula given on page 2, solve, giving your solutions as exact values

$$\cos\left(x - \frac{\pi}{6}\right) = \sin x \qquad \text{for } -\pi \leqslant x \leqslant 2\pi$$

(4)

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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24. Question 8 · part a · 2 marks · Working and answer

Working and answer

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